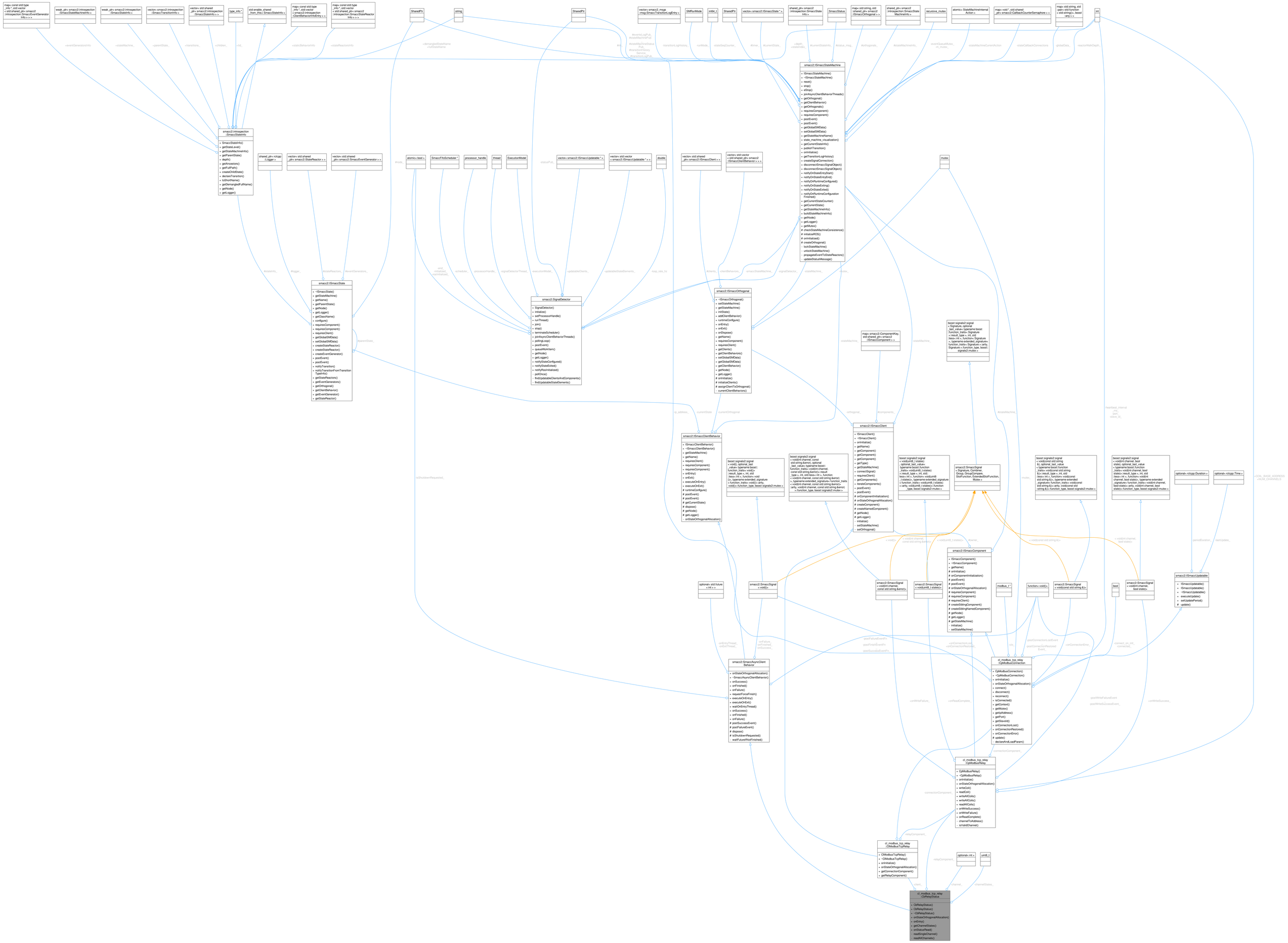
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >
weak_ptr< smacc2::introspection
::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection
::SmaccStateInfo >
vector< smacc2::introspection
::SmaccTransitionInfo >
vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >
type_info *
std::enable_shared
_from_this< SmaccStateInfo >
map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >
SharedPtr
string
SharedPtr
vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >
SMRuntMode
int64_t
SharedPtr
vector< smacc2::ISmaccState* >
shared_ptr< smacc2
::introspection::SmaccState
Info >
SmaccStatus
map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >
shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >
recursive_mutex
atomic< StateMachineInternal
Action >
map< void*, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >
int
#eventGenerators/Info_
#stateMachine_
#parentState_
#transitions_
#children_
#id_
#staticBehaviorInfo
#stateReactorsInfo
~demangledStateName
=fullStateName
#id_
#eventLogPub
#stateMachinePub
#transitionHistory
Service
#transitionLogPub
transitionLogHistory_
#runMode_
stateSeedCounter_
#timer_
#currentState_
~depth
=stateInfo*
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
#eventQueueMutex
=its_mutex_
stateMachineCurrentAction
stateCallbackConnections
globalData_
reactorWalkDepth_
smacc2::ISmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
shared_ptr< rclcpp
::Logger >
vector< std::shared
_ptr< smacc2::StateReactor > >
vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable* >
vector< std::vector
< smacc2::ISmaccUpdatable* > >
double
vector< std::shared
_ptr< smacc2::ISmaccClient > >
vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >
mutex
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfos()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyPostInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ getGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >
boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponents()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
+ setOrthogonal()
smacc2::ISmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >
boost::signals2::signal
< void(uint8_t states)
, optional_last_value<
typename boost::function
traits< void(uint8_t states)
>::result_type >, int, std
::less< int >, function< void(uint8
_t states)>, typename extended_signature
< function_traits< void(uint8_t states)
>::arity, void(uint8_t states)>::function
_type, boost::signals2::mutex >
boost::signals2::signal
< void(const std::string
&), optional_last_value
< typename boost::function
traits< void(const std::string
&)>::result_type >, int, std
::less< int >, function< void(const
std::string &)>, typename extended
_signature< function_traits< void(const
std::string &)>::arity, void(const std
::string &)>::function_type, boost::signals2::mutex >
boost::signals2::signal
< void(int channel, bool
state), optional_last_value
< typename boost::function
traits< void(int channel, bool
state)>::result_type >, int, std
::less< int >, function< void(int
channel, bool state)>, typename extended
_signature< function_traits< void(int channel,
bool state)>::arity, void(int channel, bool
state)>::function_type, boost::signals2::mutex >
optional< rclcpp::Duration >
optional< rclcpp::Time >
boost::signals2::signal
< void(), optional_last_value
< typename boost::function
traits< void()>::result_type >, int, std
::less< int >, function< void()>,
typename extended_signature< function_traits
< void()>::arity, void()>::function_type,
boost::signals2::mutex >
boost::signals2::signal
< void(int channel,
const std::string &error),
optional_last_value
< typename boost::
function traits< void(int channel,
const std::string &error)>
::result_type >, int, std
::less< int >, function
< void(int channel, const std::string &error)
>::function_type, boost::signals2::mutex >
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
optional< std::future
< int > >
smacc2::SmaccSignal
< void()>
smacc2::SmaccSignal
< void(const std::string &error)>
smacc2::SmaccSignal
< void(uint8_t states)>
modbus_t *
function< void()>
smacc2::SmaccSignal
< void(const std::string &)>
bool
smacc2::SmaccSignal
< void(int channel,
bool state)>
smacc2::SmaccAsyncClient
Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_modbus_tcp_relay
::CpModbusConnection
+ CpModbusConnection()
+ ~CpModbusConnection()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ connect()
+ disconnect()
+ reconnect()
+ isConnected()
+ getContext()
+ getMutex()
+ getIpAddress()
+ getPort()
+ getSlaveId()
+ onConnectionLost()
+ onConnectionRestored()
+ onConnectionError()
# update()
- declareAndLoadParam()
cl_modbus_tcp_relay
::CpModbusRelay
+ CpModbusRelay()
+ ~CpModbusRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ writeCoil()
+ readCoil()
+ writeAllCoils()
+ writeAllCoils()
+ readAllCoils()
+ onWriteSuccess()
+ onWriteFailure()
+ onReadComplete()
- channelToAddress()
- isValidChannel()
cl_modbus_tcp_relay
::CModbusTcpRelay
+ CModbusTcpRelay()
+ ~CModbusTcpRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ getConnectionComponent()
+ getRelayComponent()
optional< int >
uint8_t
cl_modbus_tcp_relay
::CbRelayStatus
+ CbRelayStatus()
+ CbRelayStatus()
+ ~CbRelayStatus()
+ onStateOrthogonalAllocation()
+ onEntry()
+ getChannelStates()
+ onStatusRead()
- readSingleChannel()
- readAllChannels()
#shutdownRequested_
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
#mode_
srcd_
=initialized_
=isInitialized_
=scheduler_
processorHandle_
signalDetectorThread_
executionMode_
#statusPub_
~updatableClients_
~updatableStateElements_
loop_rate_hz
#clients_
~clientBehaviors_
~smaccStateMachine_
signalDetector_
~stateMachine_
#mutex_
#stateMachine_
#orthogonal_
#components_
ip_address_
~currentState
~currentOrthogonal
connectionComponent_
relayComponent_
relayComponent_
~client
~channel_
~channelStates_
#owner_
< void()>
< void(int channel,
const std::string &error)>
< void(uint8_t states)>
< void(int channel,
bool state)>
< void(const std::string &)>
#mutex_
postFailureEventFn
postFinishEventFn
postSuccessEventFn
=onConnectionLost_
=onConnectionRestored_
=its_
=onConnectionError_
postConnectionLostEvent
postConnectionRestored
Event_
~connect_on_init_
~connected_
postWriteFailureEvent
postWriteSuccessEvent_
=onWriteSuccess_
=onWriteFailure_
=onReadComplete_
~connectionComponent_
~relayComponent_
heartbeat_interval
_ms_
~port_
~slave_id_
~period_duration_
~last_update_
=onEntryThread_
=onExitThread_
=onFailure_
=onFinished_
=onSuccess_
=GOAL_BASE_ADDRESS
=NUM_CHANNELS